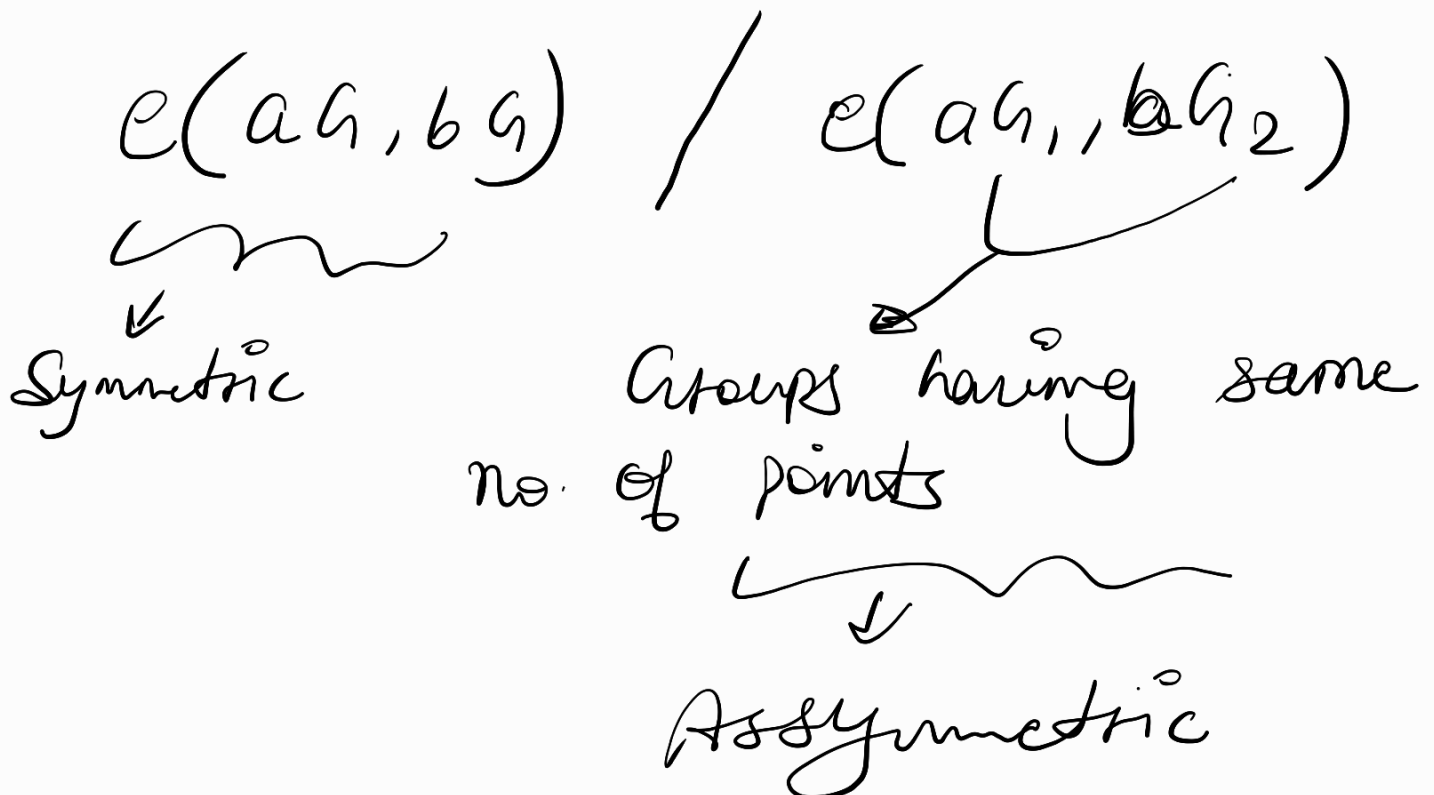


Bilinear pairings are used to prove that product of two no's is equal to a 3rd number by encrypting (not revealing) any of the three numbers.

$$E(a) \cdot E(b) = E(c); \quad E \rightarrow \text{Encryption function}$$

- Symmetric and Asymmetric groups:



Properties of bilinear pairing:

$$\begin{aligned} e(aG_1, bG_2) &= e(G_1, abG_2) \\ &= e(abG_1, G_2) \end{aligned}$$

- with regards to Ethereum, G_2 is field extension of G_1 , i.e. each element of G_2 is two dimensional so each element in G_2 has 4 coordinates

$$(x_1, x_2, y_1, y_2)$$

What does $e(a, b)$ output?

It outputs an element of another group (G_T).

The group G_T has identity element
 $(1, 0, \dots)$
12 elements.

All elements of G_T group are
12 dimensional tuples.

And G_T group is a finite cyclic
group.

NOTE: Elements in G_T behave like
powers of a base.

i.e. $(3G_2, 2G_1) = \int b^6 G_T$
3 * 2

• EIP 197:

Adds precompiles to enable
Verification of zk proofs

using bilinear pairings.

address of contract: 0×8

Q If a group is cyclic and finite, Can we always construct a homomorphism to any other finite cyclic group from the previous group?

A YES!

